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CLIMATE & SUSTAINABILITY

What the Kiln Borrows

Cement's largest emission is not burned into existence but broken out of a rock, and the rock spends the next century quietly trying to reassemble itself.

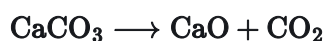
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The Carbonic Acid Entirely Expelled

Joseph Aspdin, a bricklayer from Leeds, was granted British patent 5022 on the twenty-first of October 1824 for what he called an improvement in the mode of producing an artificial stone [1]. The recipe was limestone and clay, broken into lumps and burned in something like a lime kiln. The instruction that matters is the one Aspdin gave for knowing when the burning was done: continue, he wrote, until the carbonic acid is entirely expelled.

Carbonic acid was the chemistry of the day's name for carbon dioxide. Two centuries before anyone thought of it as an emission, the defining step of making cement was described in a patent as the deliberate and complete expulsion of carbon dioxide from a rock. It is still the defining step, and it is the reason cement occupies such an awkward position in the decarbonisation of industry. Roughly seven to eight per cent of global carbon dioxide emissions come from cement [2], [3], and the larger share of that is not combustion at all.

The reaction is calcination:



Everything about cement's carbon problem follows from the three molar masses involved. Calcium carbonate is 100.086 grams per mole, quicklime 56.077, carbon dioxide 44.009. The last two sum exactly to the first, because nothing is created or lost — the kiln simply splits the rock. Every tonne of limestone decomposed releases 0.4397 tonnes of carbon dioxide; every tonne of quicklime produced carries 0.7848 tonnes of released gas behind it.

Clinker, the nodular intermediate ground up to make cement, is roughly two-thirds lime by mass. The IPCC's inventory guidelines take the lime fraction as 64.6 per cent and assume all of it came from carbonate rock, which fixes the default emission factor at $0.646 \times 0.7848 = 0.507$ tonnes of carbon dioxide per tonne of clinker [4]. That number is not an engineering estimate of a particular kiln. It is arithmetic on atomic masses, and no improvement in combustion, no alternative fuel, no electrified kiln touches it.

The scale follows in one more step. Global cement output is now around four billion tonnes a year [5], and the world's clinker-to-cement ratio can be recovered rather than looked up: dividing the reported process emissions of about 1.5 gigatonnes by four billion tonnes of cement at 0.507 gives an implied ratio near 0.74, and at 4.2 billion tonnes, 0.70. The industry's own reported figure sits inside that bracket, which is a useful check that three independently published numbers are telling the same story. The ratio has been falling — Andrew puts the global average at 0.83 in 1990 and 0.67 by 2013 [3] — and every point of decline is clinker not made and carbon not released.

A Debt, Not a Destruction

Heat is what holds the reaction open. Calcination is endothermic by about 178 kilojoules per mole, and a tonne of clinker contains roughly 11,520 moles of lime, so simply breaking the carbonate absorbs 2.05 gigajoules per tonne. The accepted theoretical minimum for making clinker is lower, near 1.75 gigajoules per tonne [6], and the gap is real chemistry rather than rounding: once the lime is free, it combines with silica and alumina to form the calcium silicates that give cement its strength, and those reactions give heat back. The kiln pays about 2.05 and is refunded about 0.30.

WHERE DOES THE REACTION WANT TO GO?

A kiln runs at 1,450 degrees Celsius to drive carbon dioxide out of limestone. At the temperature of a building site, with the atmosphere at 420 parts per million of carbon dioxide, which direction does that same reaction favour — and by how much?

Treat the decomposition with a van 't Hoff argument. Equilibrium sits where the free-energy change vanishes, so with an enthalpy of 177.8 kilojoules per mole and an entropy of 160.5 joules per mole per kelvin, the temperature at which carbon dioxide reaches one bar over the rock is their ratio, 1,108 kelvin, or about 835 degrees Celsius. The measured value is 898 degrees, and the discrepancy is honest: those are room-temperature values for quantities that drift with temperature. The shape of the answer survives the imprecision, and the shape is what matters when the pressure is not one bar but the 420 parts per million of the open air. Putting that partial pressure into the same expression moves equilibrium down to about 517 degrees Celsius.

Below that temperature, limestone is the stable phase and quicklime is not. At 298 kelvin the decomposition has a free-energy change of about **+111** kilojoules per mole against the real atmosphere, which is to say that carbonation — the reverse reaction, in which set cement takes carbon dioxide back out of the air — is downhill by that same enormous margin. Nothing in the kiln destroys the carbonate bond permanently. The kiln borrows against it, at a temperature where the loan is possible, and hands the product to a world where the repayment is thermodynamically inevitable and only kinetically slow.

In hardened cement the repayment happens through the alkaline hydration products. Portlandite, calcium hydroxide, makes up roughly a fifth of hardened paste by mass, and it reacts with dissolved carbon dioxide to regenerate calcium carbonate; when it is exhausted, the calcium silicate hydrate gel

that provides the material’s strength is attacked in turn, leaving calcium carbonate and silica gel. Mole for mole, every unit of lime that left the kiln as carbon dioxide can take a unit back.

The Front That Moves Like a Square Root

What stops the reaction from running to completion in a season is transport. Carbon dioxide must diffuse through the pore network of already-carbonated material to reach fresh alkalinity, and the reacted layer it has to cross grows as the front advances. That is the classic structure of a diffusion-limited moving boundary, and it yields the law Tuutti made the foundation of concrete durability modelling in 1982 [7]:

$$x(t) = k\sqrt{t}$$

Depth goes as the square root of time, with *k* in millimetres per root-year carrying everything else — mix design, water-to-cement ratio, curing, humidity, exposure. Sagüés and colleagues measured it directly on eighteen Florida bridges built between 1939 and 1981 and found *k* spanning zero to 14 millimetres per root-year, with a median of 1.4 and the highest values on inland bridge decks [8]. The square root is unforgiving in both directions: doubling the depth reached costs four times the wait. In years for the front to arrive:

<i>k</i> (mm yr ^{-1/2})	25 mm cover	30 mm cover	50 mm cover
1.4	319	459	1,276
3.0	69	100	278
5.0	25	36	100
8.0	10	14	39

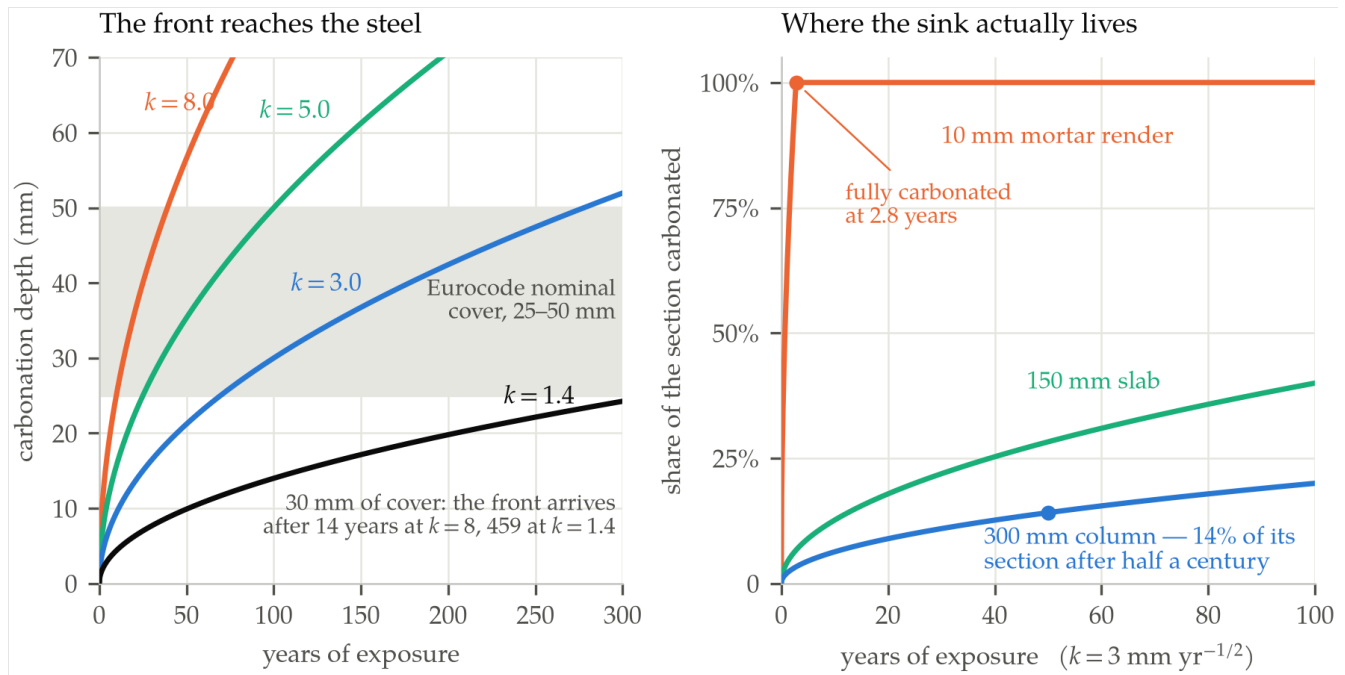


FIGURE The same one-parameter law, read two ways. Left: carbonation depth against time for four values of the rate coefficient, with the Eurocode nominal cover band shaded. Right: the carbonated share of a member of thickness d exposed on both faces, which is $2k\sqrt{t}/d$ until it saturates — a thin render is finished in under three years while a column has managed a seventh of its section in fifty.

The same law explains where the world's carbonation actually happens. For a member of thickness d attacked from both faces, the carbonated share of the section is $2k\sqrt{t}/d$ until it saturates. At a middling k of 3, a ten-millimetre mortar render is completely carbonated in 2.8 years. A 150-millimetre slab reaches 28 per cent in fifty years. A 300-millimetre column manages 14 per cent — one seventh of its cement, after half a century of standing in the weather.

Structural concrete, in other words, is a poor absorber, and thin material is an excellent one. Renders, screeds, mortar beds and above all the rubble left when a building is demolished and crushed — where a cubic metre of intact concrete becomes square metres of new surface — are where the reaction gets its opportunity. The accounting papers that estimate the global sink all partition cement use into concrete, mortar, kiln dust and construction waste for exactly this reason [9], [10].

Four Accounts of the Same Sink

The size of that sink has been estimated four times in a decade, by overlapping groups, using similar models and different data. The results do not agree.

Account	Period	Uptake (Gt CO ₂)	Offset
Xi et al., 2016 [9]	1930–2013	16.5	43%
Guo et al., 2021 [10]	1930–2019	21.0	55%
Wu et al., 2024 [11]	1930–2023	23.9	52%
Niu et al., 2025 [12]	1928–2023	21.3	46%

Xi and colleagues reported 4.5 gigatonnes of carbon, which is 16.5 gigatonnes of carbon dioxide, absorbed between 1930 and 2013 [9]. The disagreement is sharpest between the two most recent: Guo’s account reaches 21.02 gigatonnes by 2019, and Niu’s reaches 21.26 by 2023 — a quarter of a gigatonne apart across four years in which annual uptake alone should have added something like 3.2. Two careful teams, both including several of the same authors, differ by roughly the entire uptake of the interval between their endpoints.

The present-day flow shows the same spread. The Global Carbon Budget, which carries the sink as the average of two published series, puts 2023 fossil emissions at 10.1 gigatonnes of carbon with the cement sink included and 10.3 without [13] — a difference of 0.2 gigatonnes of carbon, or 0.73 gigatonnes of carbon dioxide. Wu’s account gives 0.93 for the same year [11]; Niu’s gives 0.84 [12]. Highest to lowest is a 27 per cent disagreement about a flux comparable to the annual emissions of Germany.

None of this makes the sink doubtful. Every account agrees it is large, that it has been running since the first Portland cement was set, and that its magnitude is somewhere between two-fifths and rather more than half of all cement process emissions ever made. What the spread reveals is how much of the estimate depends on assumptions no one can measure globally: how thick the world’s concrete is, how much of it has been demolished, how finely the rubble was crushed, and what happened to it afterwards.

What the Repayment Costs

There is a reason the construction industry spent a century treating this reaction as a defect rather than a service. Fresh concrete’s pore solution sits near pH 12.5 to 13, held there by portlandite, and at that alkalinity steel carries a passive oxide film that makes it effectively immune to corrosion. Carbonation consumes precisely the compound responsible. As the front passes, pH falls below 9, the film is no longer thermodynamically stable, and reinforcement that was protected becomes ordinary steel in damp salty concrete.

Tuutti’s framework splits the resulting service life into two phases: an initiation period during which the front travels through the cover and nothing visible happens, and a propagation period during which the steel corrodes, the rust expands, and the cover cracks and spalls [7]. A Spanish survey of twenty-five reinforced concrete buildings put average initiation at about 13.7 years and a further 30 years from initiation to cracking [14]. That is why Eurocode 2 specifies cover by exposure class and design life

rather than by strength alone: for the carbonation classes at a fifty-year design life, minimum durability cover runs from 25 to 50 millimetres, with a further construction tolerance on top [15]. Read the carbonation-depth table as a design chart and the logic is plain — thirty millimetres of cover buys a century at $k = 3$ and fourteen years at $k = 8$.

The conflict is not hypothetical, and it has sharpened as the sink has become attractive. A 2026 study of steel in early-age carbonated calcium sulfoaluminate-Portland mortar found that deliberately forcing carbon dioxide into fresh material accelerated corrosion sharply — mean corrosion cluster volume doubling after four hours of treatment — and named the problem outright as the carbon sinking–corrosion dilemma [16]. Every route to making the sink faster is a route to reaching the reinforcement sooner.

Two things keep the dilemma from being a crisis. The first is that the reaction is genuinely slow in good concrete: Sagüés projected 266 years to corrosion initiation for the worst decile of rate coefficients combined with the worst decile of cover in the Florida inventory, and concluded that only a small fraction of those bridges would show carbonation-induced corrosion within a 75-year life [8]. The second is that the fastest absorption happens in material with no steel in it at all — renders, mortars, crushed demolition waste. The sink and the risk are largely in different places, which is luck rather than design.

Counting a Reaction Nobody Owns

For all its size, the sink has been institutionally invisible. The 2006 IPCC inventory guidelines [4] and their 2019 refinement have no line for it, so national greenhouse gas accounts have counted cement’s process emission in full on the day the kiln ran and counted nothing at all in the decades afterwards. It is due to be addressed in a forthcoming IPCC methodology report on carbon dioxide removal expected in 2027.

Britain moved first. In its 2026 National Inventory Document the United Kingdom became the first country to report a country-specific estimate of concrete carbonation, putting the 2024 sink at 1.48 million tonnes of carbon dioxide — 29 per cent of the roughly 5.1 million tonnes of process emissions attributable to cement used in the UK, and around 0.4 per cent of territorial greenhouse gas emissions [17]. Those three figures check against each other and against an outside source: 1.48 over 5.1 is 29.0 per cent, and a sink that is 0.4 per cent of the national total implies a national total of 370 million tonnes, against the 371 million reported for 2024 [18].

A hard question sits underneath the accounting. The sink is a rebate, not a credit. Carbonation can only ever take back carbon that calcination released, and only from clinker that was actually made — so the ceiling on the sink is one hundred per cent of process emissions and not a molecule more. Against the 38.3 gigatonnes of carbon dioxide that cement production released between 1928 and 2018 [3], the 16.5 to 23.9 gigatonnes recovered to date is repayment on a debt, arriving decades late and by no means in full.

That makes the sink a poor foundation for a net-zero strategy, and a worse one the better the strategy works. Clinker substitution, the most reliable decarbonisation lever the industry has, cuts the emission and the future sink in the same stroke: lime that is never calcined cannot carbonate. A world that halves its clinker factor halves the reaction running in its walls a generation later. Only capture and storage breaks the symmetry, by keeping carbon that the rock has already given up from ever reaching the air.

Aspdin's instruction was to burn until the carbonic acid was entirely expelled. He did not mention that the stone would spend the next two centuries trying to take it back, because he had no reason to care. The reaction was always reversible; what changed is that we now have an interest in which direction it runs, and a great deal of concrete standing around quietly running it.

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